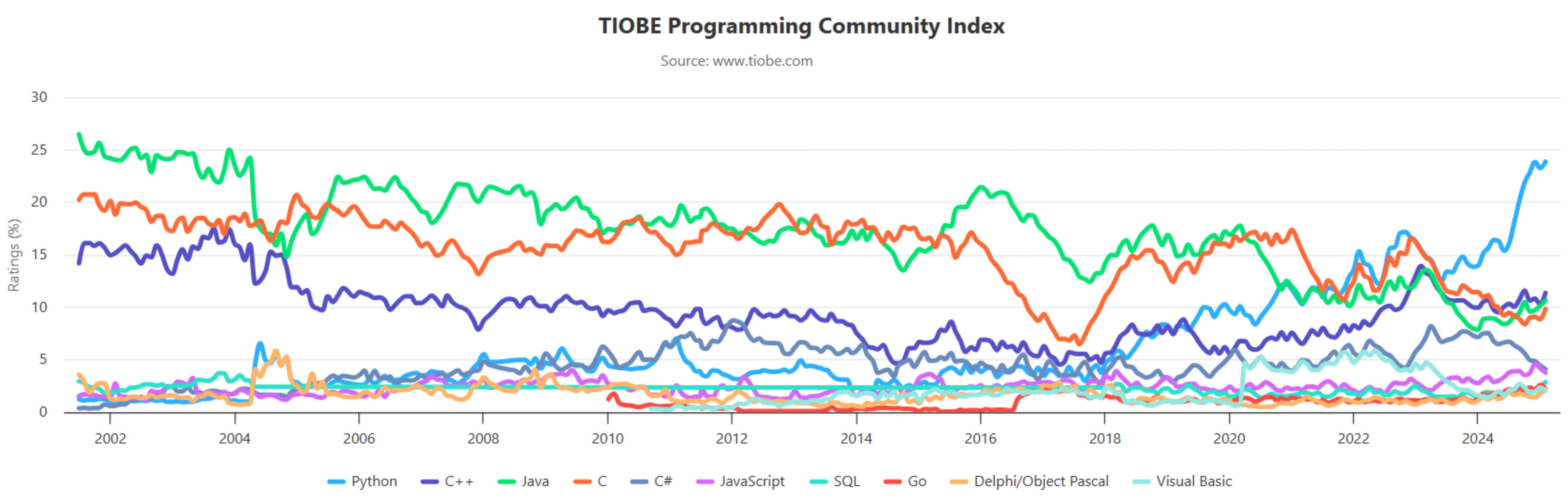
Python

- Python is a high-level, general-purpose programming language.
- Python is dynamically type-checked and garbage-collected.
- Python was conceived in the late 1980s by Guido van Rossum

The popularity of programming languages



Comments

Comments are lines of text that are ignored by the python interpreter when executing the code.

Syntax:

```
# This is a comment
```

Concatenation

Combines (concatenates) strings.

Syntax:

```
concatenated_string = string1 + string2
```

Example

Result = “Hello” + “Chandrashash”

print(Result) # HelloChandrahash

Data Types

- Integer - Float - Boolean - String

Example:

```
1.x=7
2.# Integer Value
3.y=12.4
4.# Float Value
5.is_valid = True
6.# Boolean Value
7.is_valid = False
8.# Boolean Value
9.F_Name = "John"
10.# String Value
```

Using type() print data type.

Indexing

Accesses character at a specific index.

Example:

```
My_String = “Hello”
Char = My_String[1]
print(Char) # e
```

H e l l o
0 1 2 3 4

len()

Returns the length of a string.

Syntax:

```
len(String_name)
```

Example:

```
name = “Kaushal”
Length = len(name)
print(Length) # 7
```

lower()

Converts string to lowercase.

Example:

```
Name = “Kumar”
l_name = Name.lower()
print(l_name) # kumar
```

upper()

Converts string to uppercase.

Example:

```
Name = “Kumar”
u_name = Name.uppor()
print(u_name) # KUMAR
```

print()

Prints the message or variable inside `()`.

Example:

```
print("Hello World!")    # Hello World!
print("Your name")      # Your name
```

Python Operators

- Addition (+): Adds two values together.
- Subtraction (-): Subtracts one value from another.
- Multiplication (*): Multiplies two values.
- Division (/): Divides one value by another, returns a float.
- Floor Division (//): Divides one value by another, returns the quotient as an integer.
- Modulo (%): Returns the remainder after division.

Example:

```
x = 9 y = 4
result_add = x + y # Addition   # 13
result_sub = x - y # Subtraction  # 5
result_mul = x * y # Multiplication #36
result_div = x / y # Division    # 2.5
result_fdiv = x // y # Floor Division # 2
result_mod = x % y # Modulo # 1
```

replace()

Replaces substrings.

Example:

```
my_string = "Hello"
new_text = my_string.replace("Hello", "Hi")
print(my_string) # Hello
print(new_text) # Hi
```

Slicing

Extracts a portion of the string.

Syntax:

substring = string_name[start:end]

Example:

```
my_string="Hello“
substring = my_string[0:5]
print(substring) # Hello
print(my_string[2:5]) # llo
```

split()

Splits string into a list based on a delimiter.

Example:

```
my_string="Hello"
split_text = my_string.split(",")
print(split_text) # [‘H’,’e’,’l’,’l’,’o’]
```

strip()

Removes leading/trailing whitespace.

Example:

```
my_string= "….. Hello World ….."
trimmed = my_string.strip(" . ")
print(trimmed) # Hello World
```

Variable Assignment

Assigns a value to a variable.

Syntax:

variable_name = value

Example:

Name = “Adarsh" # assigning John to variable name
x = 5 # assigning 5 to variable x

Thank you

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